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(54) **PISTON ASSEMBLY AND SPRING-LOADED CONTACT**

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H05K 7/10 (2006.01)

(52) **U.S. Cl.**

CPC **H01R 13/2421** (2013.01); **H01R 12/714** (2013.01); **H05K 7/1069** (2013.01)

(58) **Field of Classification Search**

CPC .. H01R 13/2421; H01R 12/714; H01R 13/08; H01R 4/02; H05K 7/1069

See application file for complete search history.

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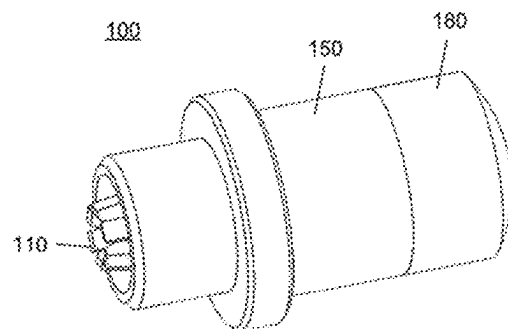
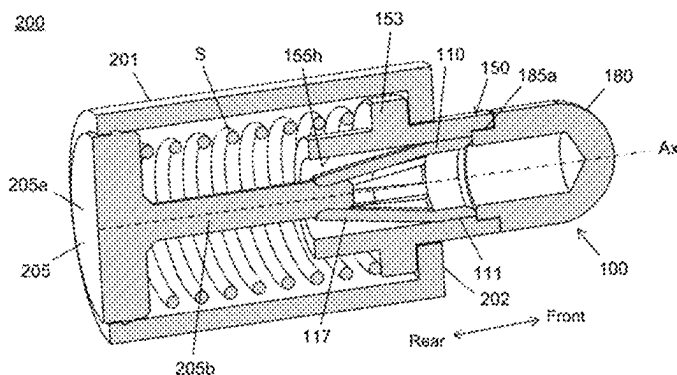
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ABSTRACT

A piston assembly for a spring-loaded contact, comprising a piston body, arranged to be received into a guiding cylinder of the spring loaded contact, a piston head, with a contacting surface arranged for contacting an external part, a clip arranged to mate with a conductive portion of the spring loaded contact, wherein the piston body comprises a through hole, the clip being arranged into the through hole, and wherein the piston head is attached to the piston body by an ultrasonic welded portion.

12 Claims, 3 Drawing Sheets



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Fig. 1

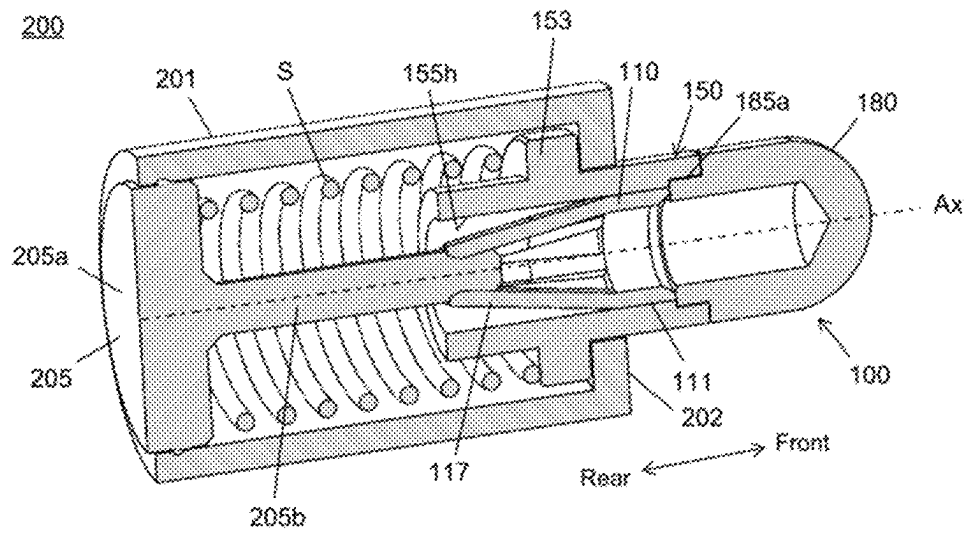


Fig. 2

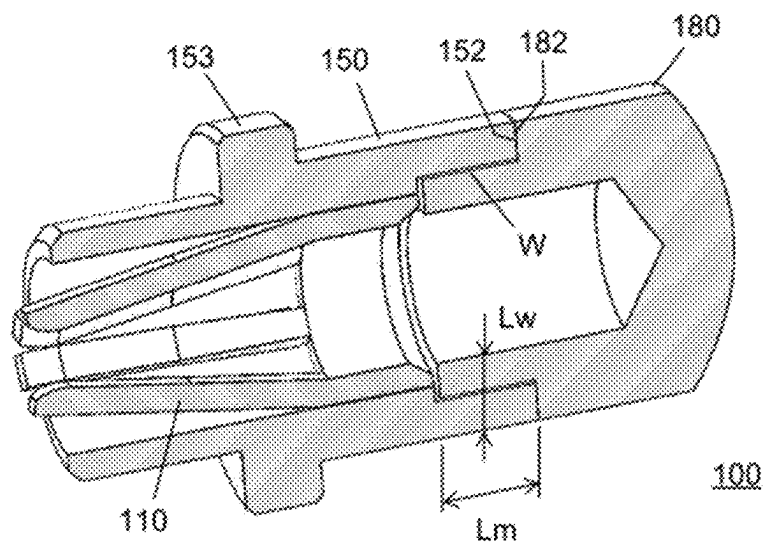


Fig. 3

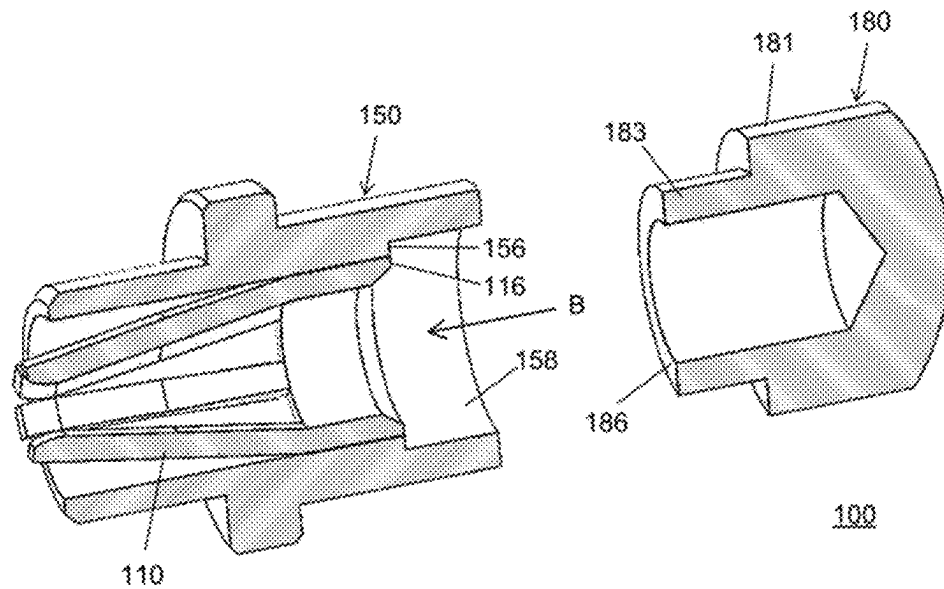


Fig. 4

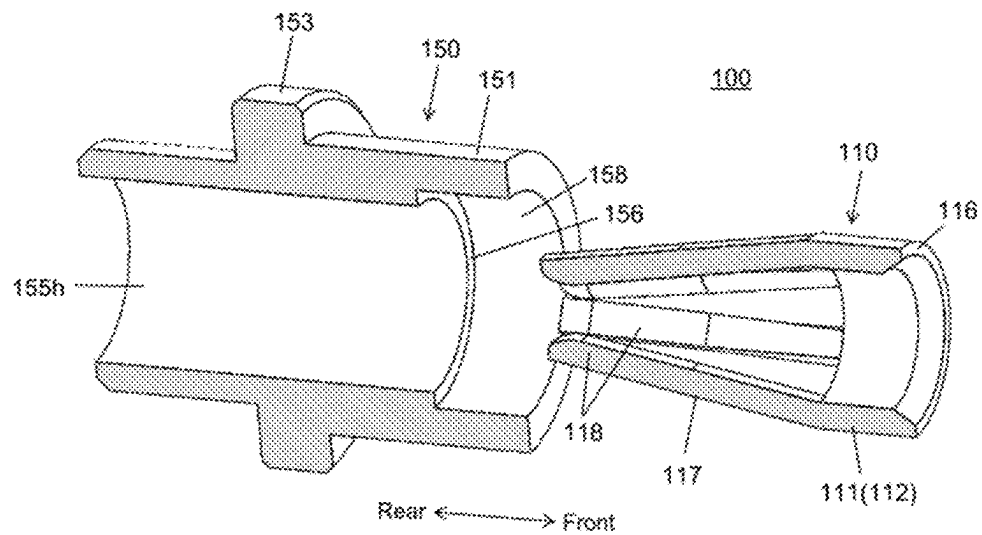


Fig. 5

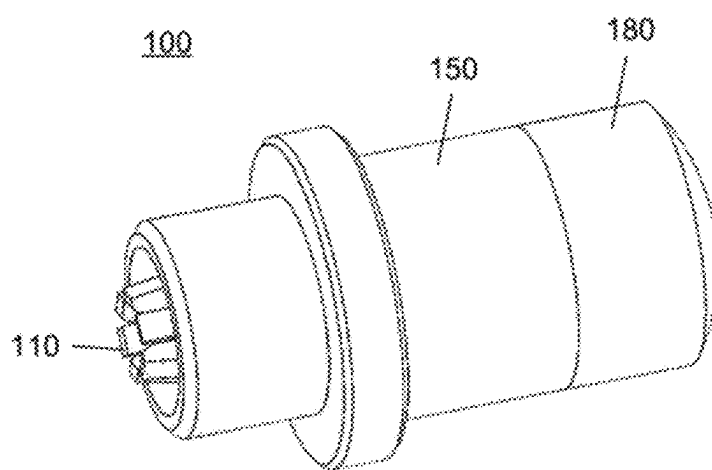
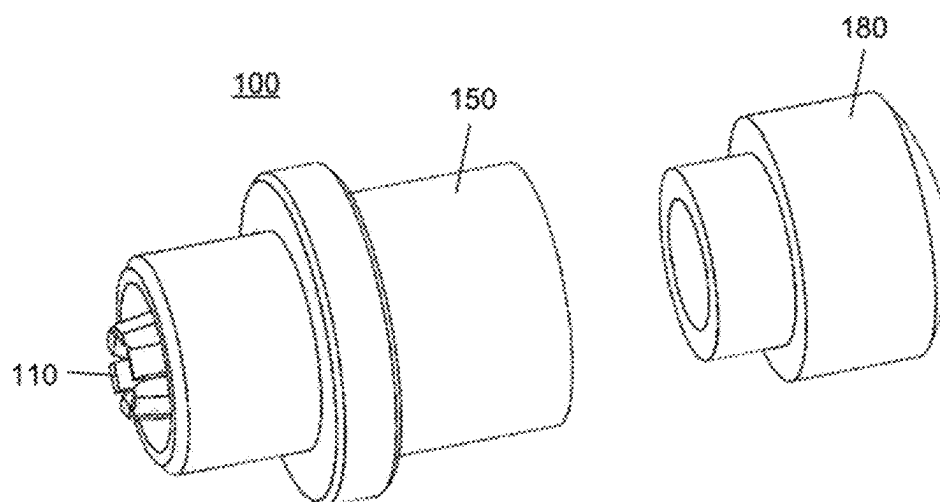


Fig. 6



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PISTON ASSEMBLY AND SPRING-LOADED CONTACT

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority to U.S. Provisional Application No. 63/282,674 filed with the United States Patent and Trademark Office on Nov. 23, 2021 and entitled "PISTON ASSEMBLY AND SPRING-LOADED CONTACT," and also claims priority to European Application No. 21169349.4 filed with the European Patent Office on Apr. 20, 2021 and entitled "PISTON ASSEMBLY AND SPRING-LOADED CONTACT," both are incorporated herein by reference in their entirety for all purposes.

DESCRIPTION

The present invention relates to a piston assembly and a spring-loaded contact.

In a spring-loaded connector which has a tubular housing hosting spring loaded piston with a clip accommodated therein with a plurality of beams, it would be conceived that the clip is inserted into the tubular housing by pushing the beams of the clip. However, this process may damage the beams and reduce a reliability of the connector.

In addition, providing a piston assembly to allow insertion of the clip by pushing it via an end face opposite to the beams requires increased wall thickness to achieve the require press fit resistance. Such increase of thickness leads to an increase of the pitch in a plug having several spring-loaded connectors.

An object of the invention is to provide a reliable compact piston assembly and a spring-loaded contact including the same.

In the above aim, a first aspect of the invention is a piston assembly for a spring-loaded contact, comprising:

- a piston body, arranged to be received into a guiding cylinder of the spring loaded contact,
- a piston head, with a contacting surface arranged for contacting an external part,
- a clip arranged to mate with a conductive portion of the spring loaded contact,
 - wherein the piston body comprises a through hole, the clip being arranged into the through hole,
 - and wherein the piston head is attached to the piston body by an ultrasonic welded portion.

According to the above structure, as the piston assembly is made of separate parts (a piston body and a piston head) and the piston head is configured to be attached to the piston body, the clip can be inserted into the through hole of the piston body from the piston head side, so that clip is not damaged during assembling. In addition, the piston head is ultrasonically welded to the piston body, thus the thickness of the piston assembly can be reduced with still providing enough strength.

Advantageously, the clip comprises:

- a flexible contacting end arranged to mate with the conductive portion of the spring-loaded contact,
- a base end, opposite to the flexible contacting end, wherein the base end is located in the through hole on a piston head side.

According to this structure, damage of the flexible contacting end is prevented when pushing the base end to insert the clip into the through hole from the piston head side.

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Advantageously, the base end comprises a tubular wall arranged to withstand a press fit force applied to the clip to insert the clip into the through hole.

Advantageously, the piston head provides an abutment to the clip. According to this structure, the clip is axially supported by the piston head.

Advantageously, the clip comprises a clipping portion opposite to the piston head.

Advantageously, at least one of the piston body, the piston head and the clip is plated before the piston head is ultrasonically welded to the piston body. One embodiment of the invention is applicable to such plated parts.

Advantageously, the welded portion is free of Heat Affected Zone, or wherein the welded portion is comprising a Heat Affected Zone having a thickness lower than 0.01 mm.

Advantageously, a wall thickness at a position where the piston body and the piston head is welded is less than 0.2 mm. Ultrasonic welding of the present embodiment can achieve this compact structure.

Advantageously, a mating length along a longitudinal axis of the piston assembly at a position where the piston head is mating with the piston body is less than 0.5 mm. Ultrasonic welding of the present embodiment can also achieve this compact structure.

Other aspect of the inventions is a spring-loaded contact, comprising:

- the piston assembly mentioned above, and
- a housing.

Advantageously, the conductive portion comprises a pin portion for mating with the clip. According to this structure, the pin portion can be securely held by the clip.

Advantageously, the conductive portion comprises a base portion for connecting an electronic or electric circuit.

Other features and advantages of the present invention will appear more clearly from the following detailed description of particular non-limitative examples of the invention, illustrated by the appended drawings where:

FIG. 1 represents a cross-sectional perspective view of a spring-loaded contact including a piston assembly.

FIG. 2 represents a cross-sectional perspective view of the piston assembly similar to that of FIG. 1.

FIG. 3 represents a cross-sectional perspective view of the piston assembly of FIG. 2, before a final assembly step.

FIG. 4 represents a cross-sectional perspective view of a piston body and a clip of the piston assembly of FIG. 2.

FIG. 5 represents a perspective view of the piston assembly of the piston assembly of FIG. 2.

FIG. 6 represents a perspective view of the piston assembly of FIG. 3.

It is of course understood that obvious improvements and/or modifications for one skilled in the art may be implemented, still being under the scope of the invention as it is defined by the appended claims.

FIG. 1 depicts one embodiment of a spring-loaded contact according to the invention and FIGS. 2 to 6 depict a piston assembly used in the spring-loaded contact.

The spring-loaded contact 200 has a cylindrical housing (guiding cylinder) 201 and a piston assembly 100 in the form of a short rod, comprising a clip 110, accommodated in the housing 201. The housing 201, piston assembly 100 and clip 110 may be designed to align concentrically with each other. The piston assembly 100 is configured to be movable in an axial direction Ax of the housing 201. In the housing 201 a coil spring S and a conductive rod 205 are disposed whose configurations and functions will be described later.

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The piston assembly **100** has a tubular piston body **150** and a cap-shaped piston head **180** attached to the end of the piston body **150** as can be seen in FIGS. 2 to 4. The clip **110** is inserted into the inside of the piston body **150**. An insertion direction is depicted with a reference character "B" in FIG. 3.

The piston body **150** and the piston head **180** can be typically machined from brass, phosphour-bronze, beryllium-copper or other copper alloy or other conductive metals/alloys, including nickel-silver alloys. The clip **110** can be manufactured from a copper alloy, typically beryllium-copper or other high strength conductive copper alloy. The parts are typically post-plated with any one of a variety of coatings, typically a gold plating, for the purpose of enhancing electrical contact and improving wear and corrosion resistance before the piston head **180** is attached to the piston body **150**.

It is noted that the shape of piston body **150** in FIG. 1 and the shape in FIGS. 2 to 6 are different, but they are not essential difference in the invention and the both embodiments can be used

The piston body **150** in FIG. 4 is provided with a tubular wall **151** having a through hole **155h** and a flange **153** formed on the outer surface of the tubular wall **151**. The flange **153** has a greater diameter than the diameter of tubular wall **151**. In the embodiment of FIG. 1, the tubular wall has a constant diameter at both sides of the flange **153**. By contrast, in the embodiment of FIGS. 2 to 6, the tubular wall **151** has a greater diameter in a front portion and a smaller diameter in a rear portion.

Inside of the piston body **150**, a receiving portion **158** is formed into which a tubular part **183** of the piston head **180** is inserted along the insert direction B. The receiving portion **158** is a cylinder-shaped space which may have a constant inner diameter greater than that of the through hole **155h**, however, in alternative embodiments, the piston body **150** might have a constant diameter through-hole. The piston body **150** further comprises an annular face **156** extending in radial direction at the end of the receiving portion **158**.

As shown in FIG. 1, the piston assembly **100** is disposed in the housing **201** such that the flange **153** is positioned within the housing **201** and a front part of the piston assembly **100** is protruding from an end wall **202** of the housing **201**. The flange **153** is pushed by way of the coil spring **S** so that the flange **153** abuts inside of the end wall **202**. The piston assembly **100** can be pushed into the housing **201** as the coil spring **S** is compressed when it contacts an external component (not shown).

The piston head **180** has a cylindrical body **181** (FIG. 3) which may have a rounded contact end. The cylindrical body **181** includes a tubular part **183** with an end annular face **186**. Outer diameter of the tubular part **183** is substantially identical to the inner diameter of the receiving portions **158** of the piston body **150** so that the surfaces can be welded to each other.

Referring to FIG. 4, the clip **110** has a base end **111** in the form of tubular wall **112** and a clipping portion **117** positioned opposite to the piston head **180**. The base end **111** is provided with an end face **116**. The base end **111** may have an outer diameter substantially identical to the inner diameter of the through hole **155h** so that the clip **110** can be pushed and inserted into the through hole **155h** (press fit assembly). The clipping portion **117** comprises a plurality of flexible beams **118** in a form of cantilever extending from the base end **111**. As each beam **118** is slightly bent radially inwardly, the diameter of the clipping portion **117** is becoming smaller toward the tip end.

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As shown in FIG. 1, the clipping portion **117** can hold the end of the conductive rod **205**. In particular, the conductive rod **205** has a circular base end **205a** and a pin portion **205b** extending vertically from the inner surface of the base end **205a**. The clipping portion **117** is configured to support the end of the pin portion **205b**. In the embodiment of FIG. 1, the circular base end **205a** can be mechanically fixed to the inside of the housing **201** (by press fit assembly, or welding assembly).

For a person skilled in the art it is understood that a different number of beams **118** can be used so long as they can hold a part of the conductive rod **205b**.

In the assembled condition, the end face **186** of the piston head **180** may abut or substantially abut the end face **116** of the clip **110** so that the clip **110** can be axially supported within the piston body **150**. Annular face **182** of the piston head **180** may also abut an end face **152** of the piston body **150**. A gap between both faces **182** and **152** can be sealed with a certain material. In a preferable embodiment, the piston head **180** and the piston body **150** may have substantially identical diameters so that there is no step at the connecting portion of the two parts.

The clip **110** is configured to be inserted into the piston body **150** along the direction B from the piston head **180**, which is in the opposite direction to a standard direction. The advantage of inserting the clip **110** in this direction is that the beams **118** of the clip **110** make contact with the rod deeper into the spring pin assembly, thus providing reliable contact in a shorter height (a length along the front-rear direction in drawings). In addition, inserting and pushing the clip **110** from the side where the piston head will be assembled provides the ability to push on end face **116**, avoiding any stress on the beams **118**.

The reverse insertion has been made possible by the assembly construction. If the piston body **150** is not a separate part, the clip **110** cannot be inserted without damaging the beams **118**. The present embodiment allows the assembler to push on the strong end of the clip **110**.

In this embodiment, the outer surface of the tubular part **183** **111** is secured to the inner surface of the receiving portion **158** of the piston body **150**. Preferably, the tubular part **183** is ultrasonic welded to a part of the piston body **150** to thereby form a circumferential welded area **W**. The welded area **W** is extending along the axial direction. The welded area **W** preferably presents a length **Lm** in axial direction less than 0.5 mm.

In addition, such ultrasonic welded area **W** is typically free of Heat Affected Zone, or comprises a very limited Heat Affected Zone having a thickness lower than 0.01 mm as temperature during ultrasonic welding remains low. The welded area **W** is also typically free of added metal.

Ultrasonic welding permits at least one of the followings: a better gold plating (especially at the bottom of holes) as the temperature does not increase during the ultrasonic welding, a better behavior in respect to the radiofrequencies, a better resistance to heat and aging due to the fact that the connector is monobloc. Further, ultrasonic welding permits that thin parts can be welded to each other. Thus, wall thickness **Lw** at a position where both parts are ultrasonic welded may be less than 0.25 mm, preferably less than 0.20 mm, and more preferably less than 0.15 mm.

In one embodiment, the annular face **182** of the piston head **180** (FIG. 2) can also be ultrasonically welded to the face **152** of the piston body **150**.

Other technologies to join two parts includes the followings: press-fitting, crimping, latching, brazing, soldering, conductive adhesives and other welding processes. How-

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ever, in press-fitting, the problem can be high force and thick walls required. In crimping, the problem can be mechanical deformation and potential plating damage. In brazing, the problem can be expensive, complicated, slow process, heat damaging the plating. In soldering, the problem can be part geometry and size, heat damaging the plating. In conductive adhesives, the problem can be low mechanical strength. In the other welding processes, the problem can be damage to plating.

According to the embodiment, the piston assembly 100 consists of separate and simple to manufacture parts; the piston body 150 and the piston head 180. This enable that the clip 110 can be inserted into the piston body 150 along the direction B, thus damaging to the beams 118 can be prevented. In addition, due to the fact that the piston head 180 is ultrasonically welded to the piston body 150, compact welded area W with sufficient reliability can be obtained.

In the present invention, both a plane rod and a hollow rod can be used for the piston head 180. In the above embodiment the piston head 180 is a male member and the piston body 150 is a female member, however, the opposite design can be used.

REFERENCE NUMERALS

100 piston assembly
111 base end
116 end face
117 clipping portion
118 flexible beam
150 piston body
151 tubular wall
153 flange
156 annular face
158 receiving portion
180 piston head
183 tubular part
201 housing
205 conductive rod
205a base end
205b pin portion
S coil spring

The invention claimed is:

1. A piston assembly for a spring-loaded contact, comprising:
 - a piston body, arranged to be received into a guiding cylinder of the spring-loaded contact,
 - a piston head, with a contacting surface arranged for contacting an external part, and
 - a clip configured to support an end of a conductive portion of the spring-loaded contact,

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wherein an end face of the piston head abuts an end face of the clip, such that the clip is axially supported within the piston body;

wherein the piston body comprises a through hole, the clip being disposed within the through hole, wherein the piston head is attached to the piston body by an ultrasonic welded portion, and wherein the spring-loaded contact further comprises a coil spring configured to move the piston assembly within the guiding cylinder.

2. The piston assembly according to claim 1, wherein the clip comprises:
 - a flexible contacting end arranged to mate with the end of the conductive portion of the spring-loaded contact,
 - a base end, opposite to the flexible contacting end,

wherein the base end is disposed within the through hole, on a piston head side.

3. The piston assembly according to claim 2, wherein the base end comprises a tubular wall arranged to withstand a press fit force applied to the clip to insert the clip into the through hole.

4. The piston assembly according to claim 1, wherein the clip comprises a clipping portion opposite to the piston head.

5. The piston assembly according to claim 1, wherein the at least one of the piston body, the piston head, or the clip is plated before the piston head is ultrasonically welded to the piston body as a welded portion.

6. The piston assembly according to claim 1, wherein the welded portion is free of Heat Affected Zone, or wherein the welded portion is comprising a Heat Affected Zone having a thickness lower than 0.01 mm.

7. The piston assembly according to claim 2, wherein a wall thickness (L_w) at a position where the piston body and the piston head is welded is less than 0.2 mm.

8. The piston assembly according to claim 2, wherein a mating length (L_m) along a longitudinal axis of the piston assembly at a position where the piston head is mating with the piston body is less than 0.5 mm.

9. A spring-loaded contact, comprising;
 - the piston assembly according to claim 1, and
 - a housing.

10. The spring-loaded contact according to claim 9, wherein the end of the conductive portion comprises a pin portion for mating with the clip.

11. The spring-loaded contact according to claim 9, wherein the conductive portion comprises a base portion for connecting an electronic or electric circuit.

12. The piston assembly according to claim 1, wherein the piston body comprises a flange that is pushed by the coil spring until the flange abuts an end wall of the guiding cylinder.

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